Kevin Cornelio Sawatzky

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Education

August 2024 – Present
Ph.D in Nuclear, Plasma & Radiological Engineering, University of Illinois Urbana-Champaign.
Graduate Concentration in Computational Science and Engineering
GPA: 4.0 / 4.0

May 2022 – April 2024
M.ASc. in Nuclear Engineering, Ontario Tech University.
Thesis title: *Coupled Radiation Transport and Mobile Depletion for Health Physics and Environmental Impact Applications.*
GPA: 4.18 / 4.3

September 2017 – April 2022
B.Eng. in Nuclear Engineering, Ontario Tech University.
Minor in Computational Science.
GPA: 3.82 / 4.3

Employment History

August 2024 – Present
Graduate Research Assistant, Nuclear, Plasma & Radiological Engineering, University of Illinois Urbana-Champaign.
- Developing adaptive mesh refinement methods and functional expansion tallies for unstructured mesh Monte Carlo particle transport.
- Developing neutronics capabilities in Cardinal to enhance advanced reactor analysis methods and workflows.

May 2026 – August 2026
Research Aide Technical - PhD, Computational Science, Argonne National Laboratory.
- Implementing hybrid Woodcock delta tracking in the OpenMC Monte Carlo neutron transport code.

May 2025 – August 2025
Research Aide Technical - PhD, Advanced Fuels and Materials, Argonne National Laboratory.
- Investigated the effects of TRISO particle homogenization on heat transport in TRISO fueled microreactors.
- Developed multi-scale methods to improve predictions of TRISO kernel temperatures during microreactor transients.

May 2024 – July 2024
Research Assistant, Engineering Physics, McMaster University.
- Developed methodologies to analyze surrogate TRISO fuel particles with HR(S)TEM and FIB-SEM instruments at the Canadian Center for Electron Microscopy.
- Co-authored a journal manuscript on high-resolution microscopy methods for TRISO fuel.

Employment History (continued)

May 2022 – April 2024

Graduate Research Assistant, Department of Energy and Nuclear Engineering, Ontario Tech University.
- Developed a discrete ordinates radiation transport solver and finite volume fluid activation/depletion solver in the MOOSE framework.
- Coupled both solvers together for the purposes of analyzing fluid depletion in the health physics and environmental impact code CARIBOU.
- Managed high performance computing resources within the Atkinson lab.

September 2022 – December 2023

Teaching Assistant, Department of Energy and Nuclear Engineering, Ontario Tech University.
Courses taught: *Nuclear Plant Operations, Nuclear Reactor Kinetics, Medical Imaging*

May 2021 – August 2021

Undergraduate Research Assistant, Department of Energy and Nuclear Engineering, Ontario Tech University.
Project title: *Literature Review of Small Modular Reactor Instrumentation and Control Strategies*.
- Performed a literature review on Small Modular Reactor instrumentation and control strategies with a focus on integral Pressurized Water Reactors and High Temperature Gas-Cooled Reactors.
- Assisted in preparing a manuscript for journal submission on the results of the literature review.

May 2020 – August 2020

Undergraduate Research Assistant, Department of Energy and Nuclear Engineering, Ontario Tech University.
Project title: *Development of a Radioisotope Atmospheric Dispersion Model in the MOOSE Finite Element Framework*.
- Performed a literature review on methods used for modeling radionuclide dispersion.
- Developed a MOOSE mini-application to perform atmospheric dispersion calculations, simulating the first 24 hours of the Chernobyl plume.

Research Interests

Radiation Transport Methods — Neutron transport, charged particle transport, discrete ordinates methods, Monte Carlo methods and variance reduction, code verification and validation.

Multi-Physics Methods — Coupled neutronics, radionuclide depletion and thermal-hydraulics; multi-physics verification and validation.

Advanced Reactor Simulation — Modeling pebble bed, molten salt, liquid metal cooled, and prismatic gas-cooled reactors.

Publications

Journal Articles

1. **K. C. Sawatzky**, A. J. Novak, and K. D. Atkinson, "A Self-Adjoint Scalar Flux Equation for Calculating Point Source Uncollided Flux Moments Without Ray Effects," *Nuclear Science and Engineering*, 2025.
DOI: https://doi.org/10.1080/00295639.2025.2592177.

Technical Reports

1. Y. Miao, S. Lee, **K. Sawatzky**, *et al.*, "Multiphysics Modeling of Microreactors with NEAMS codes, and Validation Based on KRUSTY Reactivity Insertion," Argonne National Laboratory, ANL/NEAMS-25/2, 2025.
2. B. Wirth, A. Novak, E. Merzari, W. Brinkley, and **K. Sawatzky**, "Developing a Science-based Predictive Simulation Capability for the Design and Analysis of Advanced Nuclear Fuel Concepts Operating in a Virtual Reactor Environment," Sawtooth Simulation, DE-SC0025110, 2025.

Conference Proceedings

1. E. W. Ahammed, **K. Sawatzky**, L. Gross, *et al.*, "Development of Mesh Tally Amalgamation Algorithm for Coupled High Fidelity Multiphysics Simulation," in *Proceedings of PHYSOR 2026*, Turin, Italy: Politecnico di Torino, 2026.
2. M. A. Mendes, M. Eltawila, **K. C. Sawatzky**, A. J. Novak, and R. C. Martineau, "Validation of Cardinal Coupled Neutronics and Thermomechanics Using the KRUSTY Experiments," in *Proceedings of NETS 2026*, Dayton, Ohio, USA: American Nuclear Society, 2026.
3. C. H. H. Moore, **K. C. Sawatzky**, A. J. Novak, and K. D. Atkinson, "Validation of a Two-Step Neutronics Workflow using Gnat and OpenMC with the ZED-2 Research Reactor," in *Proceedings of PHYSOR 2026*, Turin, Italy: Politecnico di Torino, 2026.
4. **K. C. Sawatzky**, E. W. Ahammed, P. P. Wilson, P. C. Shriwise, P. K. Romano, and A. J. Novak, "Mesh Tally Adaptive Mesh Refinement Strategies for Spatial Gradients Driven by Material Heterogeneities," in *Proceedings of PHYSOR 2026*, Turin, Italy: Politecnico di Torino, 2026.
5. **K. C. Sawatzky**, A. J. Novak, and Y. Miao, "Application of a Multiscale Temperature Treatment to Multiphysics Calculations in a TRISO Fueled Microreactor," in *Proceedings of PHYSOR 2026*, Turin, Italy: Politecnico di Torino, 2026.
6. **K. C. Sawatzky**, E. W. Ahammed, P. P. Wilson, P. C. Shriwise, P. K. Romano, and A. J. Novak, "Adaptive Mesh Refinement Applied to Unstructured Mesh Tallies in Cardinal-OpenMC," in *Proceedings of ANS-2025*, Chicago, Illinois, U.S.A: American Nuclear Society, 2025.
7. **K. C. Sawatzky** and A. J. Novak, "Heuristic-Based Adaptive Mesh Refinement Applied to Ray Effect Mitigation Methods," in *Proceedings of M&C-2025*, Denver, Colorado, U.S.A: American Nuclear Society, 2025.
8. **K. Sawatzky** and K. D. Atkinson, "Verification of a CARIBOU-OpenMC Workflow for the Analysis of HTGR-Like Systems Using the Proposed Ontario Tech Subcritical Assembly," in *Proceedings of PHYSOR-2024*, San Francisco, California, U.S.A: American Nuclear Society, 2024.
9. **K. Sawatzky** and K. D. Atkinson, "Development of a Near-Field Neutron Activation and Dispersion Solver for the Environmental Impact Code CARIBOU," in *Proceedings of M&C-2023*, Niagara Falls, Ontario, Canada: Canadian Nuclear Society, 2023.

Skills

Programming Languages	C, C++, Python, LaTeX, Java, MATLAB, FORTRAN
Frameworks	MOOSE, FEniCS
Tools	OpenMC, Cardinal, Siemens NX, coreform Cubit, DRAGON3/5
Miscellaneous	ParaView, git, MS Office suite